

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN PHYSIOGRAPHY

MODULE - 1

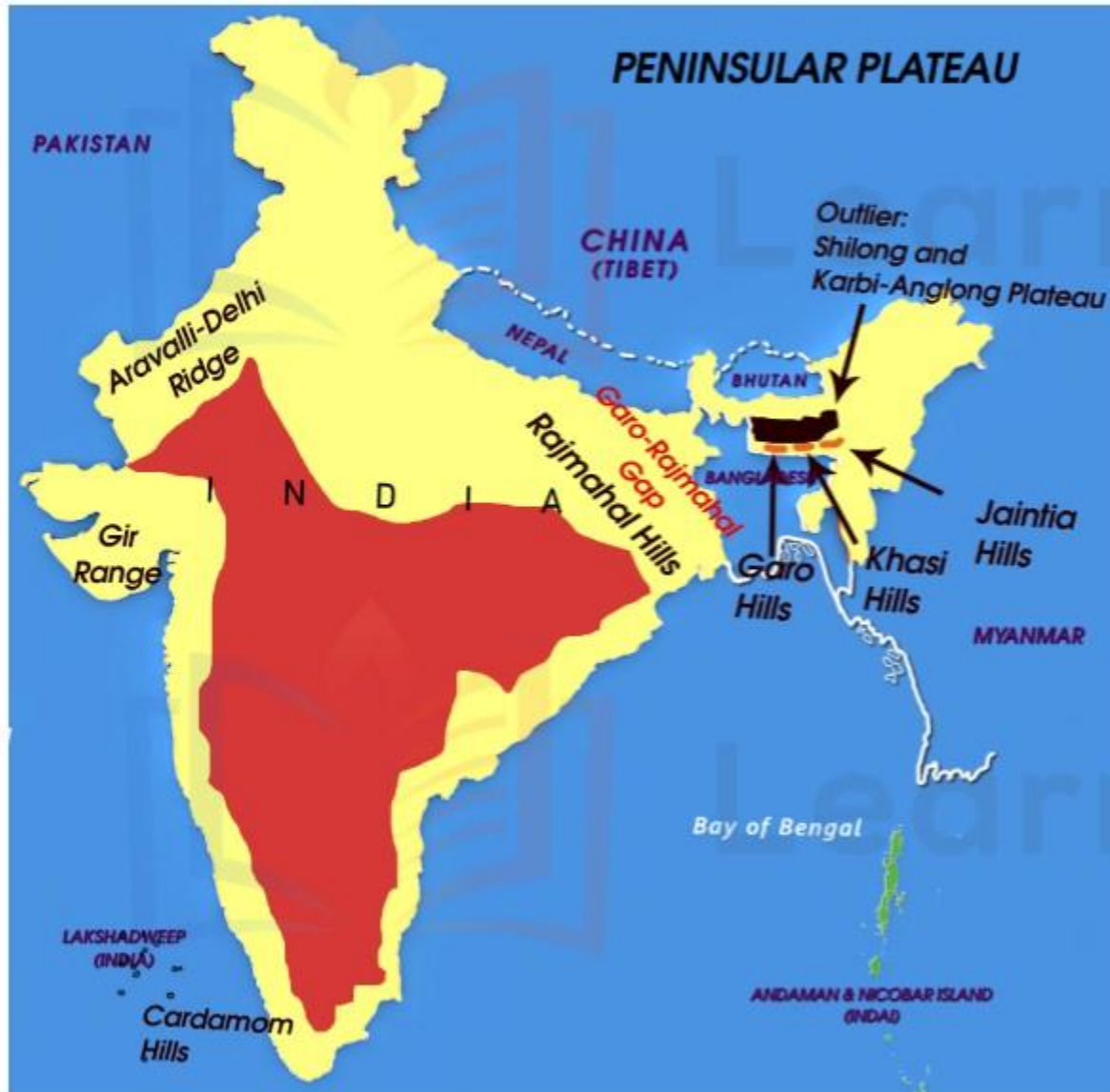
PHYSIOGRAPHIC DIVISIONS

- **HIMALAYAN RANGES**

PHYSIOGRAPHIC DIVISIONS OF INDIA

- 1** 'Physiography' of an area is the outcome of structure, process and the stage of development.
- 2** The land of India is characterised by great diversity in its physical features.
- 3** The north has a vast expanse of rugged topography consisting of a series of mountain ranges with varied peaks, beautiful valleys and deep gorges.
- 4** The south consists of stable table land with highly dissected plateaus, and denuded rocks. In between these two lies the vast north Indian plain.
- 5** Based on these macro variations, India can be divided into the following physiographic divisions:
 1. The Northern and Northeastern Mountains
 2. The Northern Plain
 3. The Peninsular Plateau
 4. The Indian Desert
 5. The Coastal Plains
 6. The Islands.

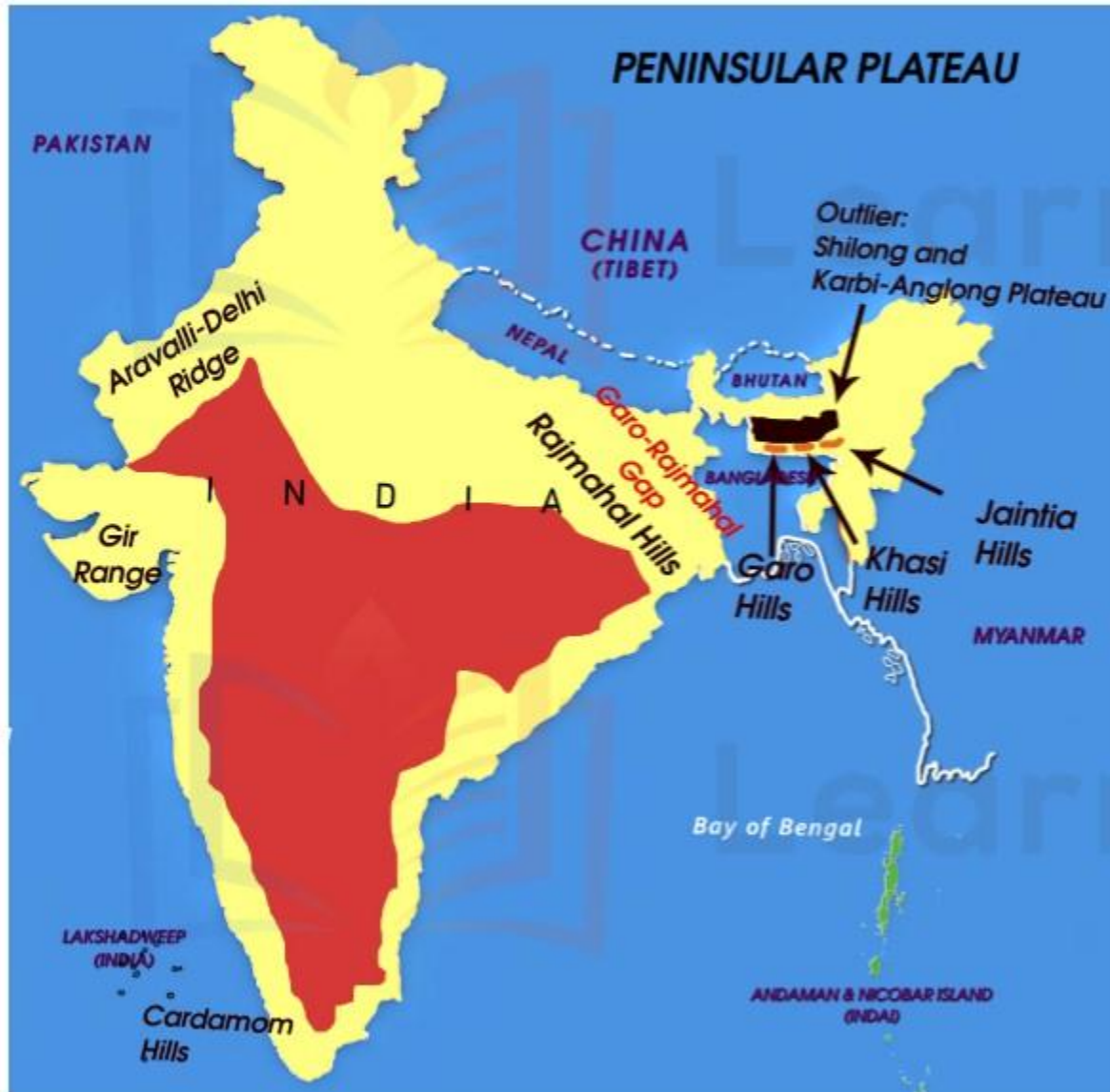
THE PENINSULAR BLOCK



1 The northern boundary of the Peninsular Block may be taken as an irregular line running from Kachchh along the western flank of the **Aravali Range** and then roughly parallel to the Yamuna and the Ganga as far as the **Rajmahal Hills** and the Ganga delta.

2 Apart from these, the **Karbi Anglong** and the **Meghalaya Plateau** in the northeast and Rajasthan in the west are also extensions of this block.

THE PENINSULAR BLOCK



3 The north-eastern parts are separated by the **Malda fault in West Bengal** from the Chotanagpur plateau.

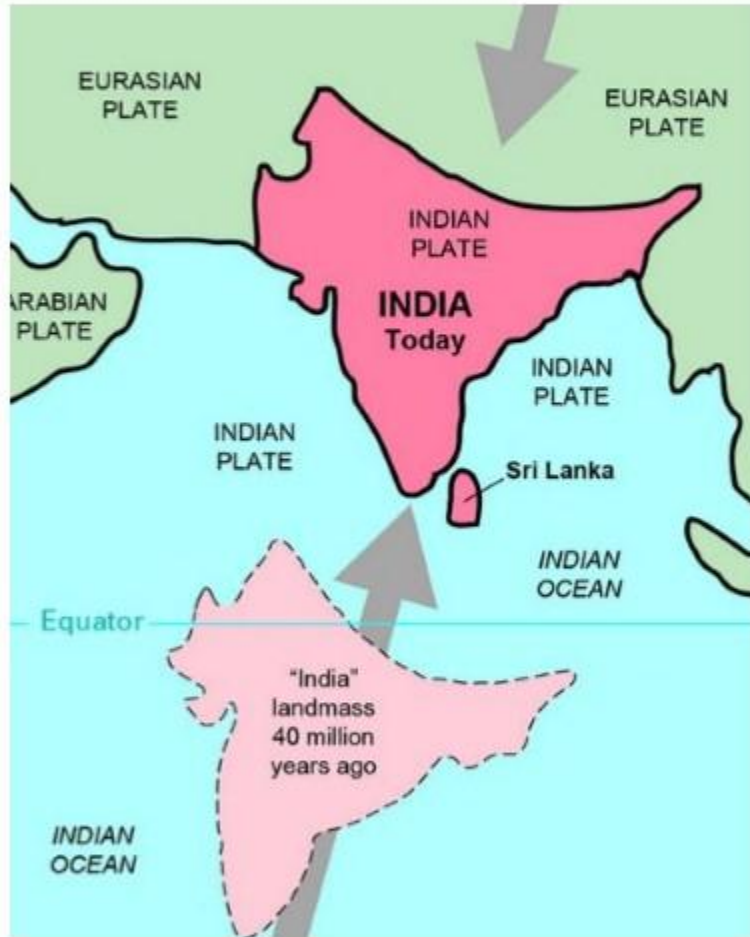
4 In Rajasthan, the **desert and other desert-like features** overlay this block.

5 The Peninsula is formed essentially by a great complex of very ancient gneisses and granites, which constitutes a major part of it.

PENINSULAR PLATEAU



THE PENINSULAR BLOCK



***Movement of Indian Plate
northwards***

6

Since the Cambrian period, the Peninsula has been standing like a rigid block with the exception of some of its western coast which is submerged beneath the sea and some other parts changed due to tectonic activity **without affecting the original basement.**

7

As a part of the Indo-Australian Plate, it has been subjected to various vertical movements and block faulting.

8

The rift valleys of the Narmada, the Tapi and the Mahanadi and the Satpura block mountains are some examples of it.

THE PENINSULAR BLOCK



9 The Peninsula mostly consists of relict and residual mountains like the Aravali hills, the Nallamala hills, the Javadi hills, the Veliconda hills, the Palkonda range and the Mahendragiri hills, etc.

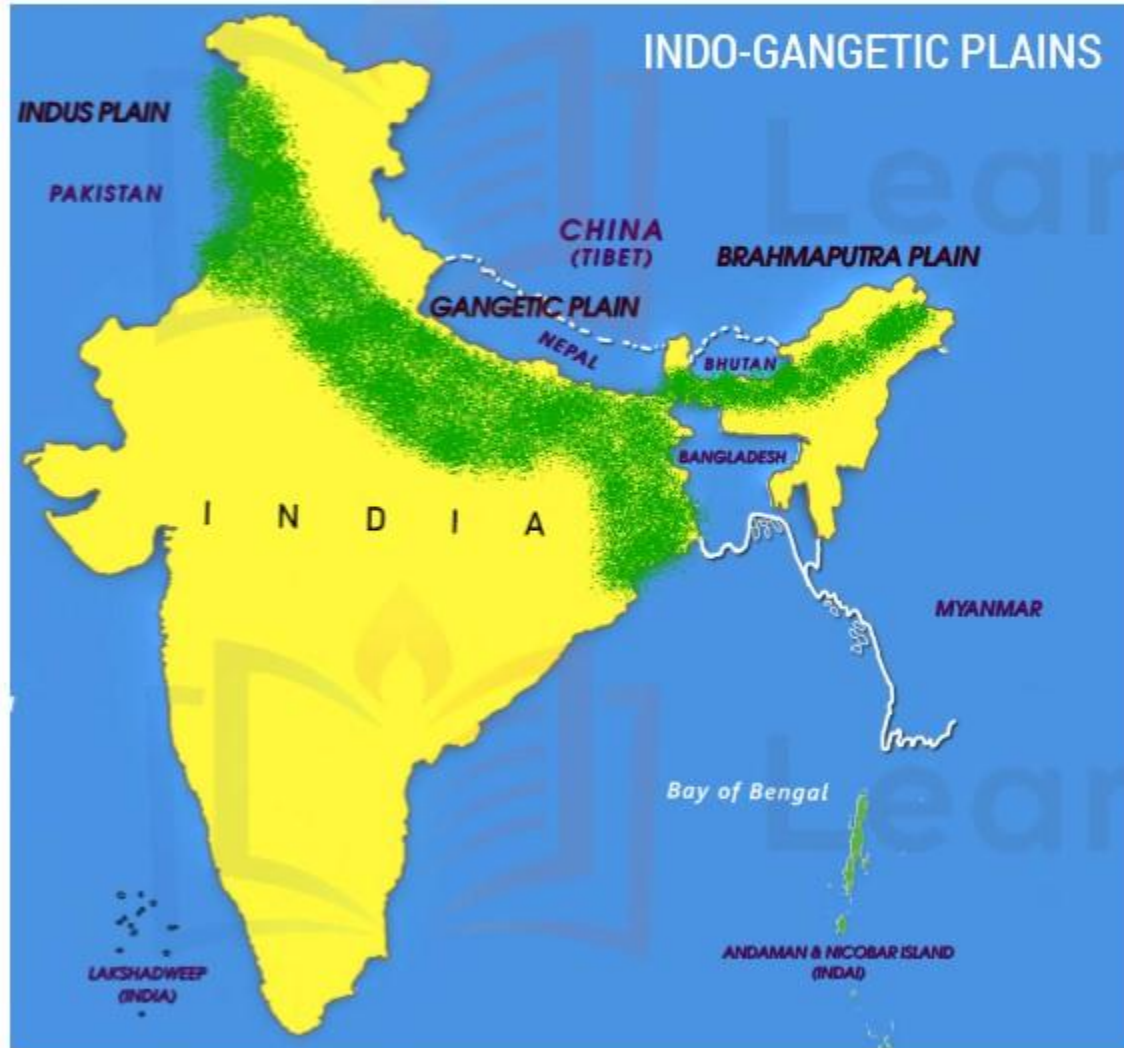
10 The river valleys here are shallow with low gradients.

THE HIMALAYAN MOUNTAIN RANGES



- 1** The Himalayas are young, weak and flexible in their geological structure unlike the rigid and stable Peninsular Block.
- 2** Consequently, they are still subjected to the interplay of exogenic and endogenic forces, resulting in the development of faults, folds etc.
- 3** These mountains are tectonic in origin, dissected by fast-flowing rivers which are in their youthful stage.
- 4** Various landforms like gorges, V-shaped valleys, rapids, waterfalls, etc. are indicative of this stage.

INDO-GANGA-BRAHMAPUTRA PLAIN



- 1** The third geological division of India comprises the plains formed by the river Indus, the Ganga and the Brahmaputra.
- 2** This plain attained its maximum development during the third phase of the Himalayan mountain formation approximately about 64million years ago.
- 3** Since then, it has been gradually filled by the sediments brought by the Himalayan and Peninsular rivers.

WHAT CAN WE CONCLUDE FROM THIS?



- It is evident from the above discussion that there are significant variations among the different regions of India in terms of their geological structure, which has far-reaching impact upon other related aspects.
- Variations in the physiography and relief are important among these.
- The relief and physiography of India has been greatly influenced by the geological and geomorphological processes active in the Indian subcontinent.

INDIA

PAKISTAN

CHINA
(TIBET)

NEPAL

BHUTAN

BANGLADESH

MYANMAR

Bay of Bengal

LAKSHADWEEP
(INDIA)

ANDAMAN & NICOBAR ISLAND
(INDIA)

MAJOR PHYSIOGRAPHIC DIVISION OF INDIA

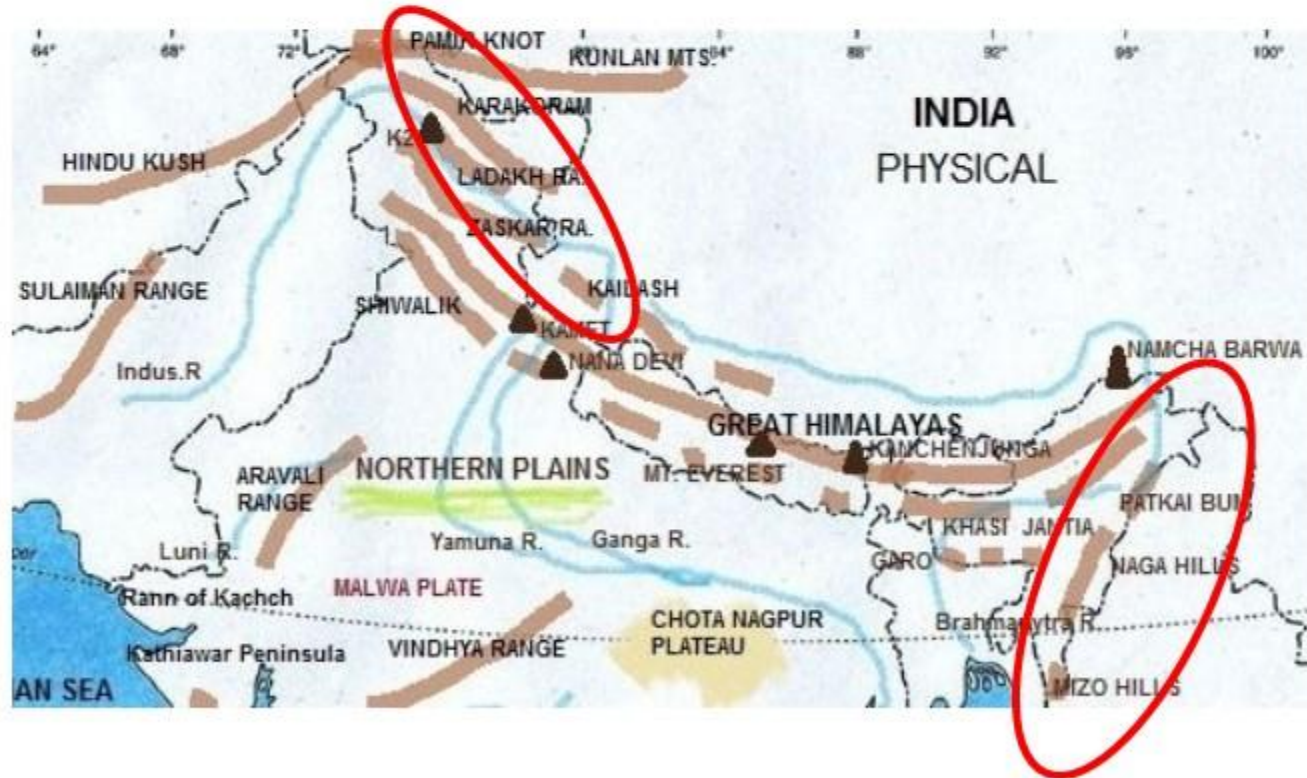
1. The Northern and Northeastern Mountains
2. The Northern Plains
3. The Thar Desert
4. The Peninsular Plateau
5. The Coastal Plains
6. The Islands

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THE NORTHERN AND NORTHEASTERN MOUNTAINS

1

HIMALAYAN RANGES



Geographically the entire Himalayan region can be divided into three ranges

1. The Himalayan ranges
 - a) Shiwalik or Outer Himalayas
 - b) Lesser or Middle Himalayas
 - c) The Greater Himalayas
2. Trans Himalayas
3. Eastern hills or Purvanchal

THE NORTHERN AND NORTHEASTERN MOUNTAINS

1

1

HIMALAYAN RANGES

- 1 This comprises of a series of several parallel or converging ranges.
- 2 The ranges are separated by deep valleys **creating a highly dissected topography.**
- 3 The southern slopes have steep gradients and northern slopes have comparatively gentler slopes.
- 4 Most of the Himalayan ranges fall in India, Nepal and Bhutan.
- 5 The northern slopes are partly situated in Tibet, while the western extremity lies in Pakistan, Afghanistan and Central Asia.
- 6 Himalayas between Tibet and Ganga plain is actually a succession of three parallel ranges.

THE NORTHERN AND NORTHEASTERN MOUNTAINS

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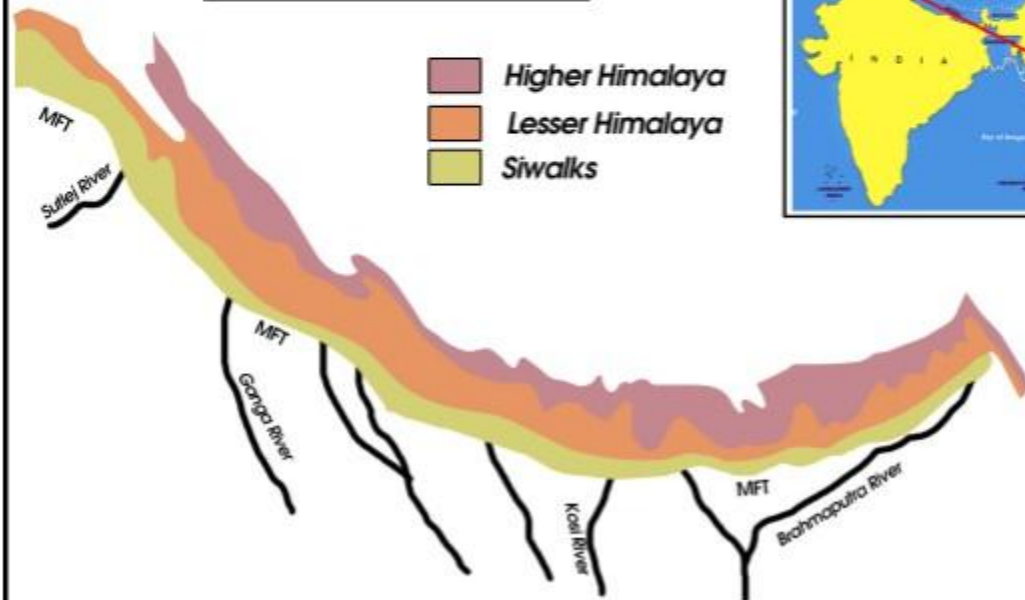
HIMALAYAN RANGES

a

SHIWALIK RANGES

DIVISION OF THE HIMALAYAS

-  Higher Himalaya
-  Lesser Himalaya
-  Siwalks



1

This is also known as **Outer Himalayas**.

2

It is located between the **Great Plains** and **Lesser Himalayas**.

3

The altitude varies from 600 to 1,500 metres

4

It runs almost parallel to the lesser Himalayas for a distance of about 2,400 kms from the **Potwar Plateau** to the **Brahmaputra Valley**

INDIAN PHYSIOGRAPHY

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HIMALAYAN RANGES

a

SHIWALIK RANGES

- 5 The width of the Shiwaliks vary from 50 km in Himachal Pradesh to less than 15 km in Arunachal Pradesh
- 6 They are almost unbroken chain of low hills except for a gap of 80 to 90 km, which is occupied the valley of the Tista River
- 7 Shiwaliks from Northeast India up to Nepal are covered with thick forest, but, the forest cover becomes thin in the west.
- 8 The southern slopes of this range are almost completely devoid of forest cover in Punjab and Himachal Pradesh as they get highly dissected by several seasonal streams called Chos.
- 9 Shiwaliks are formed last of all the ranges. They are formed due to intense folding and hardened due to compression because of the northward movement of the Indian plate.

INDIAN PHYSIOGRAPHY

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HIMALAYAN RANGES

a

SHIWALIK RANGES

HOW DOES FORMATION OF DUNS OCCUR IN SHIWALIKS?



Dehradun

1

As the Shiwalik hills were formed after the formation of main Himalayas, *they obstructed the course of the rivers draining from the higher reaches of Himalayas.*

2

This led to the formation of temporary lakes, and the debris brought by those rivers deposited in these lakes after the rivers had cut their course through Shiwalik ranges, the lakes were drained away leaving behind *plains called Duns or Doons.* Dehradun in Uttarkhand is the best example.

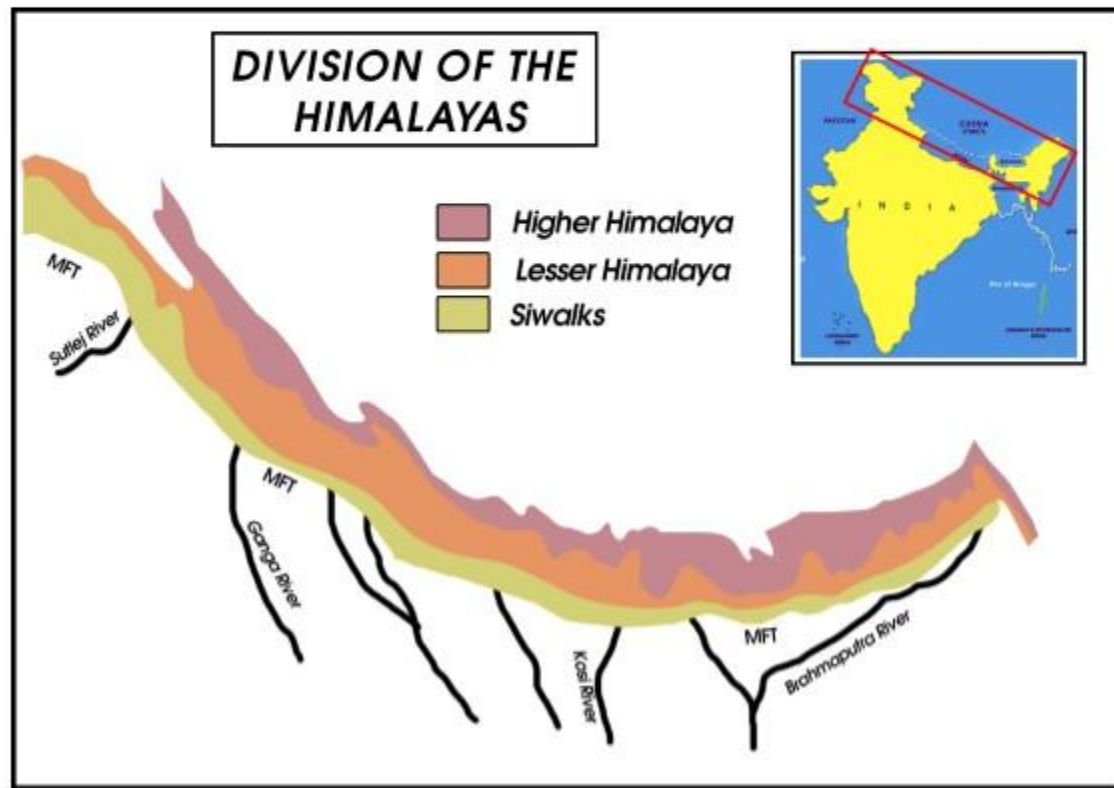
INDIAN PHYSIOGRAPHY

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HIMALAYAN RANGES

b

LESSER HIMALAYAS



1

In between the Shiwaliks in the south and the Great Himalayas in the north is the Middle Himalayas running almost parallel to both the ranges

2

It is also called the Himachal or Lower Himalayas.

3

Middle Himalayas are 60 to 80 kms wide and elevation ranges from 3,500 kms to 4,500 kms above the sea level. They are 2,400 km in length.

INDIAN PHYSIOGRAPHY

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HIMALAYAN RANGES

b

LESSER HIMALAYAS



IMPORTANT RANGES FROM LESSER HIMALAYAS

REGION

The Pir Panjal Range

Jammu and Kashmir

The Dhauladhar Range

Himachal Pradesh

The Mussoorie Range and The Nag Tiba Range

Uttarakhand

Mahabharat Lekh

Nepal



INDIAN PHYSIOGRAPHY

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HIMALAYAN RANGES

b

LESSER HIMALAYAS



4

The Pir Panjal range in Kashmir is the longest and the most important range.

5

It extends from the Jhelum river to the upper Beas river for 300 to 400 km and it is separated from Zaskar Range by the valley of Kashmir.

6

Another significant valley of this region is the Kangra valley in Himachal Pradesh.

7

Majority of the Himalayan hill resorts like Shimla, Mussoorie, Ranikhet, Nainital, Almora and Darjeeling, etc. are located in this regions.

FAMOUS HILL RESORTS OF HIMACHAL



INDIAN PHYSIOGRAPHY

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HIMALAYAN RANGES

C

GREATER HIMALAYAS



1

This is also known as the **inner Himalayas**, or **Himadri ranges**.

2

Average elevation is about 6,100 meters above sea level and it has an average width of about 25 km.

3

It is mainly formed of crystalline granites and gneisses.

4

The folds in this range are asymmetrical and **the mountain arc is convex to the south** terminated abruptly in the Nanga Parbat in northwest and in the Namcha Barwa in the Northeast.



INDIAN PHYSIOGRAPHY

1

HIMALAYAN RANGES

C

GREATER HIMALAYAS

REGIONAL NAME OF MOUNT EVEREST	REGION
Sagarmatha (The Goddess of the Sky)	Nepal
Chomlungma (Mother of the World)	Tibet



5

This mountain range boasts of the tallest peaks of the world, most of which remain under perpetual snow.

6

Mount Everest was first located by George Everest, the then Surveyor General of India in 1841 and in 1852 it was **established as the highest peak of the world by the Great Trigonometrical Survey of India.**

7

The other important peaks in descending order of altitudes are Kanchenjunga, Lhotse, Makalu, Dhaulagiri etc.

1 HIMALAYAN RANGES

WHAT DO YOU UNDERSTAND BY SYNTAXIAL BENDS OF HIMALAYAS?



SYNTAXIAL BENDS OF HIMALAYAS:

- Himalayan ranges take sharp southward bends at Indus and Brahmaputra gorges in the west and east respectively.
- These bends are called **Syntaxial Bends of the Himalayas**.
- The western syntaxial bend occurs near the Nanga Parbat, where the Indus river cuts a deep gorge.
- The eastern syntaxial bend occurs near the Namche Barwa, where the Brahmaputra river cuts the deep gorge.

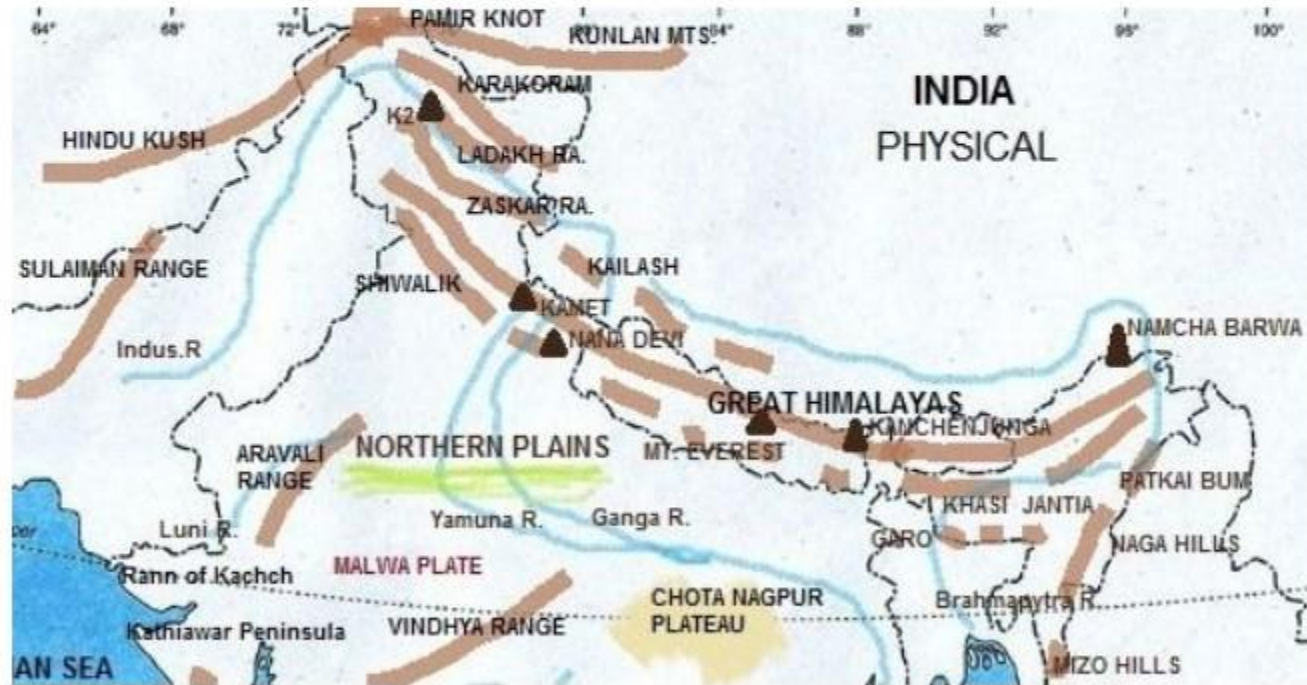


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